

Paired input: RGB (3 ch) + IR (1 ch)

Training-only feature gating
VIS 20% | IR 20% | VIS+IR 60%
An absent backbone is not executed

RGB backbone
YOLOv10-s, layers 0-10

IR backbone
YOLOv10-s, 1-channel

RGB P3

IR P3

L4 | 128 ch | s=8

Additive or FAM
P3 fusion

fused
P3

RGB P4

IR P4

L6 | 256 ch | s=16

Additive or FAM
P4 fusion

fused
P4

RGB P5

IR P5

L10 | 512 ch | s=32

Additive or FAM
P5 fusion

fused
P5

Shared FPN / PAN neck
Top-down: P5 → P4 → P3 | Bottom-up: P3 → P4 → P5

v10Detect
one-to-many +
one-to-one

Person
detections

Standalone routes

VIS+IR [1,1]: both backbones; FAM active

VIS [1,0]: RGB only; FAM bypassed

IR [0,1]: IR only; FAM bypassed